# ViLD: Open-Vocabulary Object Detection

Gu, Lin, Kuo, Cui — Google Research & Nvidia. Distilling VLMs for open-vocab detection.

## Task Overview

- Detect and name objects from an extensible vocabulary, including categories never seen in training.
- Open-set recognition: map any region to a text label using a shared, robust vision–language embedding space.
- Class-agnostic localization: propose high-quality boxes or masks without relying on fixed classes.



## OV Gaps

Data scarcity: base labels miss the long tail

- Recog-loc gap; decouple via distillation
- Efficiency: many regions vs large label sets




## OV Classifiers

- CLIP/ALIGN learn image–text alignment, enabling open-vocabulary recognition.
- Work on cropped regions via prompt–image similarity; localization is external.
- Distill their recognition into detectors to approach end-to-end open-vocabulary detection.

# OV Detectors

- Detectors predict boxes and labels jointly, often with modest vocabularies or via dataset transfer.
- Zero-shot and open-vocabulary detectors boost novel-class recall yet still trail supervised AP.
- Naive class-to-detection transfer falls short; tighter localization–recognition coupling needed.



# Pairwise Fusion

- Text embeddings: rich semantics and on-the-fly vocab.
- Image embeddings: visual regularities for strong localization.
- ViLD fuses both: text-supervised + image-distilled for robust, open-vocab detection.



# ViLD Framework

- Teacher: frozen VLM; outputs text embeddings for prompts and image embeddings for region crops
- Student: detector proposes regions; aligns features to prompts; at test scores regions vs prompts
- Objectives: CE on sims p(c| r)=softmax(sim(fr,tc)); L1 aligns detector features to teacher crops



# Prelims

Region r → e_r from multi-scale crops, then L2-normalize.

- Score e_r vs T(a_j); softmax over z/τ gives prompt probs.
- Train ViLD-text XEnt on base prompts; then score + NMS.

$$\mathcal{V}(\text{crop}(I, \tilde{r}_{\{1\times, 1.5\times\}})) = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \text{ where } \mathbf{v} = \mathcal{V}(\text{crop}(I, \tilde{r}_{1\times})) + \mathcal{V}(\text{crop}(I, \tilde{r}_{1.5\times})). \tag{1}$$

$$\begin{aligned} \mathbf{e}_r &= \mathcal{R}(\phi(I), r) \\ \mathbf{z}(r) &= \left[sim(\mathbf{e}_r, \mathbf{e}_{bg}),\ sim(\mathbf{e}_r, \mathbf{t}_1),\ \cdots,\ sim(\mathbf{e}_r, \mathbf{t}_{|C_B|})\right] \\ \mathcal{L}_{\text{ViLD-text}} &= \frac{1}{N} \sum_{r \in P} \mathcal{L}_{\text{CE}}\Big(softmax(\mathbf{z}(r)/\tau), y_r\Big), \end{aligned} \tag{2}$$

- Notation
- Scoring

# ViLD: 3 Phases

- Phase 1: Class-agnostic localization — proposals + masks that generalize beyond base classes.
- Phase 2 Recognition: ViLD-text aligns regions to prompts, ViLD-image distills from cropped regions.
- Phase 3: Ensembling — combine text and image scores with rule that respects base vs novel labels.



# Phase 1

Detect class-agnostic regions and masks via proposals.

- Classify regions by cosine similarity to text prompts.
- Train with temp-softmax CE on base prompts; infer any text.

$$\Pr(v_i \mid \mathbf{e}_r) = softmax_i(sim(\mathbf{e}_r, \mathcal{T}(\mathbf{v}))/\tau), \tag{7}$$

$$\Pr(a_j \mid \mathbf{e}_r) = softmax_j(sim(\mathbf{e}_r, \mathcal{T}(\mathbf{a}))/\tau). \tag{8}$$

- Proposals
- ViLD-text



# KD + Ens

ViLD-image: L1 align region features to teacher image embeddings from the same crops

- Multi-scale crops stabilize regions; complements ViLD-text semantics with visual structure
- Train: L = L_text + w*L_image. Infer: weighted geo-mean; base=detector*text, novel=text-led

$$\mathcal{L}_{\text{ViLD}} = \mathcal{L}_{\text{ViLD-text}} + w \cdot \mathcal{L}_{\text{ViLD-image}}, \tag{4}$$

$$p_{i,\text{ensemble}} = \begin{cases} p_{i,\text{ViLD-text}}^{\lambda} \cdot p_{i,\text{cls}}^{(1-\lambda)}, & \text{if } i \in C_B \\ p_{i,\text{ViLD-text}}^{(1-\lambda)} \cdot p_{i,\text{cls}}^{\lambda}, & \text{if } i \in C_N \end{cases} \tag{5}$$

- Visual KD
- Ensembles



# Results

LVIS AP strong; COCO competitive; big rare-class gains.

- ViLD generalizes across LVIS, COCO, and Objects365.
- Apps: expand vocab via prompts; interactive; code+demo ready.

- Takeaways: split locate/recognize; text; distill; ensemble.

| Backbone | Method | Box $AP_r$ | Box $AP_c$ | Box $AP_f$ | Box AP | Mask $AP_r$ | Mask $AP_c$ | Mask $AP_f$ | Mask AP |
|---|---|---|---|---|---|---|---|---|---|
| ResNet-50 +ViT-B/32 | CLIP on cropped regions | 19.5 | 19.7 | 17.0 | 18.6 | 18.9 | 18.8 | 16.0 | 17.7 |
| | ViLD-text+CLIP | **23.8** | 26.7 | 32.8 | 28.6 | **22.6** | 24.8 | 29.2 | 26.1 |
| ResNet-50 | Supervised-RFS (base+novel) | 13.0 | 26.7 | 37.4 | 28.5 | 12.3 | 24.3 | 32.4 | 25.4 |
| | GloVe baseline | 3.2 | 22.0 | 34.9 | 23.8 | 3.0 | 20.1 | 30.4 | 21.2 |
| | ViLD-text | 10.6 | 26.1 | 37.4 | 27.9 | 10.1 | 23.9 | 32.5 | 24.9 |
| | ViLD-image | 10.3 | 11.5 | 11.1 | 11.2 | 11.2 | 11.3 | 11.1 | 11.2 |
| | ViLD ($w$=0.5) | 16.3 | 21.2 | 31.6 | 24.4 | 16.1 | 20.0 | 28.3 | 22.5 |
| | ViLD-ensemble ($w$=0.5) | **16.7** | 26.5 | 34.2 | 27.8 | **16.6** | 24.6 | 30.3 | 25.5 |
| ResNet-152 | Supervised-RFS (base+novel) | 16.2 | 29.6 | 39.7 | 31.2 | 14.4 | 26.8 | 34.2 | 27.6 |
| | ViLD-text | 12.3 | 28.3 | 39.7 | 30.0 | 11.7 | 25.8 | 34.4 | 26.7 |
| | ViLD-image | 12.5 | 13.9 | 13.4 | 13.4 | 13.1 | 13.4 | 13.0 | 13.2 |
| | ViLD ($w$=1.0) | 19.1 | 22.4 | 31.5 | 25.4 | **18.7** | 21.1 | 28.4 | 23.6 |
| | ViLD-ensemble ($w$=2.0) | **19.8** | 27.1 | 34.5 | 28.7 | **18.7** | 24.9 | 30.6 | 26.0 |
| EfficientNet-b7 | ViLD-ensemble w/ ViT-L/14 ($w$=1.0) | 22.0 | 31.5 | 38.0 | 32.4 | 21.7 | 29.1 | 33.6 | 29.6 |
| | ViLD-ensemble w/ ALIGN ($w$=1.0) | **27.0** | 29.4 | 36.5 | 31.8 | **26.3** | 27.2 | 32.9 | 29.3 |


- LVIS AP +
- COCO+O365